

Estimation Error in Portfolio Optimisation

Bayesian Shrinkage, CVaR & Wasserstein Distributional Robustness

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1 Introduction

Modern portfolio theory begins with the idea that investment decisions should be made at the portfolio level rather than by considering securities individually. Markowitz (1952) formalised this by choosing portfolio weights based on a trade-off between expected return and risk.

The difficulty is that the quantities needed to construct the “optimal” portfolio are not actually known. If r_t is the vector of asset returns, then we would ideally know the true expected return

$$\mu = E[r_t]$$

and covariance matrix

$$\Sigma = \text{Cov}(r_t).$$

In reality, these have to be estimated from historical data:

$$\hat{\mu} = \frac{1}{T} \sum_{t=1}^T r_t, \quad \hat{\Sigma} = \frac{1}{T-1} \sum_{t=1}^T (r_t - \hat{\mu})(r_t - \hat{\mu})^\top.$$

The problem is that these estimates are noisy. An industry can have a high historical average return simply because it had a good five years, rather than because its true expected return is genuinely higher. An optimiser can take small differences like this very seriously and turn them into large portfolio positions.

This project looks at four increasingly sophisticated ways of dealing with that problem, using equal weighting as a benchmark:

1. Classical mean-variance optimisation using the raw sample estimates.
2. Mean-variance optimisation using shrinkage to make those estimates more stable.
3. CVaR optimisation, which changes the definition of risk from general volatility to severe losses.
4. Wasserstein robust CVaR, which also allows for the possibility that the historical return distribution is not exactly the distribution we will see in the future.

The main question is whether each additional layer actually improves the portfolio once it is tested on data that was not used to construct it.

2 Data and Backtesting

I use the twelve value-weighted US industry portfolios from the Kenneth R. French Data Library, covering areas such as Energy, Manufacturing, Business Equipment, Healthcare, Utilities and Money. Monthly data run from January 2010 to September 2025.

The portfolio is updated once a month. For every month being tested, only the previous 60 months are available. The first 48 months are used to fit candidate models and the final 12 months are used to choose parameters such as the level of risk aversion. The model is then refitted using the full 60 months and the portfolio weights are fixed before the next month’s return is revealed.

The window then moves forward by one month and the process repeats. This gives 129 out-of-sample monthly returns from January 2015 to September 2025. No test-month return is used to choose the portfolio that is evaluated on that month.

All portfolios are long-only, fully invested, and no industry can make up more than 35% of the portfolio. I also charge 10 basis points for each unit of one-way portfolio turnover so that a strategy cannot trade constantly without any cost.

3 The Portfolio Methods

3.1 Equal Weighting

The benchmark is simply

$$w_i = \frac{1}{12}.$$

Each industry receives 8.33% of the portfolio. This ignores expected returns and correlations entirely. That sounds unsophisticated, but it is useful precisely because it makes almost no estimation assumptions. If a complicated optimisation method cannot beat a diversified portfolio that requires no forecasts, that is important information.

3.2 Sample Mean-Variance Optimisation

The classical Markowitz portfolio solves

$$\min_w \left\{ \frac{\gamma}{2} w^\top \hat{\Sigma} w - \hat{\mu}^\top w \right\},$$

subject to

$$\sum_i w_i = 1, \quad 0 \leq w_i \leq 0.35.$$

The first term penalises risk and the second rewards estimated return. The parameter γ determines how much weight is given to risk relative to return.

The weakness is that $\hat{\mu}$ and $\hat{\Sigma}$ are treated as if they were reliable. If one industry happens to have had a particularly strong five years, the optimiser can allocate heavily towards it. Similarly, an unusually favourable historical correlation can make a combination of assets look safer than it really is. Optimisation therefore has the potential to amplify estimation error rather than remove it.

3.3 Shrinkage Mean-Variance Optimisation

The next method keeps exactly the same Markowitz optimisation problem but changes the estimates being fed into it. This makes it a useful comparison: if performance improves, the improvement is coming from better estimation rather than from changing the portfolio objective.

Expected returns are shrunk towards the cross-sectional average:

$$\mu_i^B = B_i m_i + (1 - B_i) m_0, \quad B_i = \frac{\tau^2}{\tau^2 + v_i/T}.$$

Here, m_i is the observed mean return for industry i and m_0 is the average mean return across industries. If an individual estimate is noisy, more weight is placed on the overall average. If it appears more reliable, more of the original estimate is retained.

The covariance matrix is also regularised using Ledoit–Wolf shrinkage:

$$\hat{\Sigma}^{LW} = \delta F + (1 - \delta) \hat{\Sigma}.$$

The raw covariance matrix is blended with a simpler and more stable target matrix F . The idea is to give up some detail in exchange for estimates that are less sensitive to one particular historical sample.

The new portfolio therefore solves the same Markowitz problem, but using $\hat{\mu}^B$ and $\hat{\Sigma}^{LW}$ instead of the raw sample estimates.

3.4 Historical CVaR

The next change is more fundamental. Mean-variance optimisation treats unexpectedly high and unexpectedly low returns as the same type of risk because both increase variance. For an investor, however, a large positive return is not normally a problem.

Conditional Value-at-Risk (CVaR), or Expected Shortfall, focuses directly on losses. At the 95% level, CVaR is the average loss among the worst 5% of outcomes. If portfolio loss is

$$L_t = -w^\top r_t,$$

historical CVaR can be minimised using

$$\min_{w, \eta, u} \left\{ \eta + \frac{1}{(1 - \alpha)T} \sum_{t=1}^T u_t \right\},$$

where $\alpha = 0.95$ and the u_t variables measure losses beyond the threshold η .

I also require the portfolio's shrunk expected return to be at least as high as the estimated return of the equal-weight portfolio. Otherwise, reducing CVaR could simply mean moving into a very low-return portfolio.

This method therefore keeps the improved expected-return estimate from the shrinkage model, but changes what the optimiser means by "risk".

3.5 Wasserstein Robust CVaR

Historical CVaR has another weakness. It assumes that the 60 historical monthly returns are a good representation of the true return distribution. At a 95% confidence level, however, only about

$$60(0.05) = 3$$

observations sit in the relevant tail. That is not much information to base a downside-risk estimate on.

The Wasserstein approach allows the true return distribution to lie within a small neighbourhood of the historical one:

$$\mathcal{P} = \{P : W_1(P, \hat{P}) \leq \epsilon\}.$$

Rather than minimising CVaR under exactly the observed historical distribution, the portfolio is chosen to perform well under the least favourable nearby distribution.

For the formulation used here, this gives

$$\text{CVaR}^{\text{robust}} = \text{CVaR}^{\text{historical}} + \frac{\epsilon}{1 - \alpha} \|w\|_\infty.$$

Since

$$\|w\|_\infty = \max_i w_i,$$

the extra term penalises the largest individual portfolio weight. This is intuitive: a portfolio that puts 35% into one industry is more vulnerable to being wrong about that industry's return distribution than a more evenly spread portfolio.

So the progression from one method to the next is fairly direct: first improve the estimates, then improve the definition of downside risk, and finally account for uncertainty in the historical distribution itself.

4 Results

The main results are:

Strategy	Return	Vol.	Sharpe	Max DD
Equal Weight	11.43%	15.62%	0.773	-23.22%
Sample MVO	7.63%	16.51%	0.528	-29.62%
Shrinkage MVO	8.11%	12.32%	0.697	-19.30%
Historical Mean-CVaR	9.49%	13.24%	0.753	-21.14%
Wasserstein Robust CVaR	9.93%	13.63%	0.766	-23.29%

The first thing that stands out is how badly the raw Markowitz portfolio performed. It earned only 7.63% per year, had the highest volatility and had the deepest drawdown. It also changed its mind constantly: average monthly turnover was 22.0%, compared with only 1.3% for equal weighting.

Shrinkage made a large difference. It reduced volatility from 16.51% to 12.32%, improved maximum drawdown from -29.62% to -19.30%, and cut monthly turnover from 22.0% to 4.9%. This suggests that a large part of the problem with sample MVO came from unstable inputs rather than from the basic Markowitz idea itself.

The CVaR methods sat somewhere between equal weighting and shrinkage MVO. Historical CVaR returned 9.49% with a Sharpe ratio of 0.753, while Wasserstein robust CVaR returned 9.93% with a Sharpe of 0.766. The robust portfolio had the best Sharpe ratio of the four optimisation methods, although equal weighting still had the highest return and a slightly higher Sharpe.

The effect of the Wasserstein penalty is easier to see by looking at concentration:

Strategy	Turnover	Effective N	Largest Weight
Equal Weight	1.3%	12.0	8.3%
Sample MVO	22.0%	3.5	34.4%
Shrinkage MVO	4.9%	4.0	34.4%
Historical CVaR	9.6%	3.6	35.0%
Robust CVaR	10.2%	4.8	24.4%

Sample MVO, shrinkage MVO and historical CVaR all regularly pushed their largest position towards the 35% cap. Wasserstein CVaR reduced the average largest position to 24.4% and increased the effective number of industries from 3.6 under historical CVaR to 4.8. That is exactly what the extra robustness penalty is supposed to do.

There was also no single method that dominated in every difficult market period. Historical CVaR performed best during the COVID sell-off, losing 18.0% compared with 22.7% for equal weighting. During the 2022 tightening period, however, shrinkage MVO did considerably better, losing 8.2% compared with 18.8% for Wasserstein CVaR. Robustness to one type of uncertainty does not mean protection against every market regime.

5 How Certain Are the Results?

One problem with any backtest is that the exact historical period matters. The fact that one strategy beats another between 2015 and 2025 does not necessarily mean that it has a permanently higher expected return.

To get some idea of this uncertainty, I used a moving-block bootstrap. Six-month sections of the out-of-sample return history were sampled with replacement and joined together to create 2,000 alternative return histories. Using blocks rather than individual months preserves some of the short-term dependence in returns.

For Wasserstein CVaR, the estimated annual return difference relative to equal weighting was -1.7 percentage points, with a 95% bootstrap interval of

$$[-4.7\%, 1.1\%].$$

The Sharpe-ratio difference was approximately -0.01, with an interval of

$$[-0.21, 0.18].$$

Both intervals include zero. Therefore, there is not enough evidence to say that the robust strategy is genuinely better than equal weighting. This does not mean the two strategies are necessarily equal; it means the backtest does not establish a reliable advantage.

6 Conclusion

The most useful result from the project is not that one method “won”. In fact, the simplest portfolio—equal weighting—had the highest compound return. What changed as the methods became more so-

phisticated was mainly the way risk and uncertainty were handled.

Raw mean-variance optimisation was extremely sensitive to the historical estimates. It produced concentrated portfolios, high turnover and poor out-of-sample performance. Shrinking the expected returns and covariance matrix made the same Markowitz framework much more stable and produced the lowest volatility and best maximum drawdown.

CVaR then changed the focus from general volatility to bad downside outcomes. Wasserstein CVaR went one step further by recognising that even the historical distribution itself is uncertain. The resulting penalty produced a noticeably less concentrated portfolio.

None of these changes solved the hardest part of portfolio construction: knowing which assets will actually deliver the highest future returns. The more sophisticated methods were useful mainly because they stopped uncertain estimates from turning into overly confident portfolio decisions.

The overall progression can therefore be summarised as

$$\text{Sample MVO} \longrightarrow \text{Shrinkage MVO} \longrightarrow \text{CVaR} \longrightarrow \text{Robust CVaR},$$

where each step addresses a different weakness in the one before it: noisy parameters, an imperfect measure of risk, and finally uncertainty about the historical return distribution itself.

References

- [1] H. Markowitz, “Portfolio Selection,” *The Journal of Finance*, Vol. 7, No. 1, pp. 77–91, 1952.
- [2] O. Ledoit and M. Wolf, “A Well-Conditioned Estimator for Large-Dimensional Covariance Matrices,” *Journal of Multivariate Analysis*, Vol. 88, No. 2, pp. 365–411, 2004.
- [3] R. T. Rockafellar and S. Uryasev, “Optimization of Conditional Value-at-Risk,” *Journal of Risk*, Vol. 2, No. 3, pp. 21–41, 2000.
- [4] P. Mohajerin Esfahani and D. Kuhn, “Data-Driven Distributionally Robust Optimization Using the Wasserstein Metric,” *Mathematical Programming*, Vol. 171, pp. 115–166, 2018.
- [5] V. DeMiguel, L. Garlappi and R. Uppal, “Optimal Versus Naive Diversification: How Inefficient is the $1/N$ Portfolio Strategy?” *Review of Financial Studies*, Vol. 22, No. 5, pp. 1915–1953, 2009.
- [6] K. R. French, “12 Industry Portfolios,” Kenneth R. French Data Library, Dartmouth College.